NEET UG • PHYSICS CORE SERIES

Moving Charges and Magnetism

Comprehensive Theory, Rigorous Derivations, Magnetic Field Rules, Device Principles, and NEET-Pattern Solved Assessment.

1. Overview & NEET UG Examination Blueprint

In Class 12 Electrostatics, electric charges at rest were seen to establish an electric field $\mathbf{E}$. In this foundational module of Electromagnetism, we explore how **charges in motion** (steady currents) establish a **magnetic field** $\mathbf{B}$ in the surrounding space, how external fields exert forces on moving charges and current elements, and how these interactions are engineered into precision instruments like the cyclotron and the moving coil galvanometer.

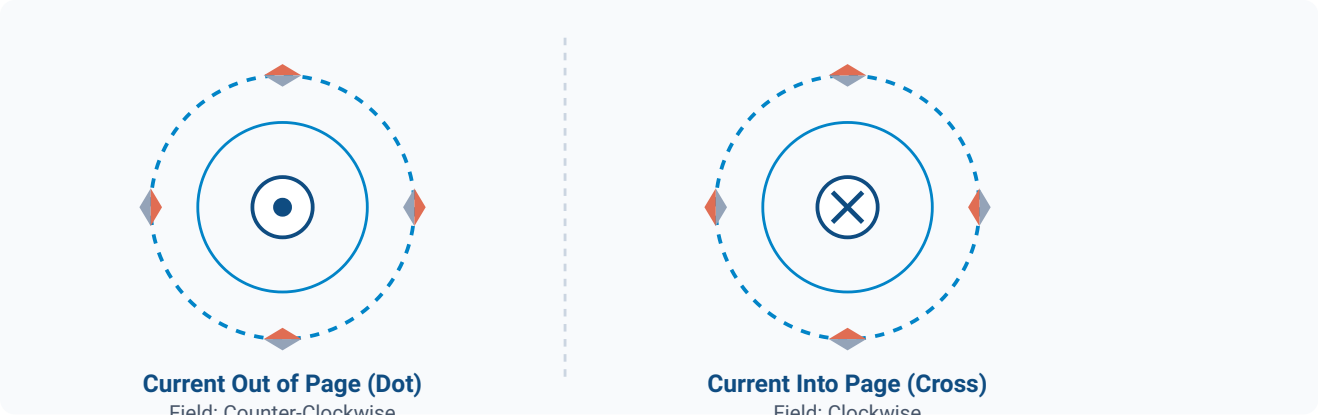
> **NEET UG Weightage & High-Yield Topic Index**
>
> Consistently accounts for **2 to 3 questions (8 – 12 marks)** every year. The highest frequency concepts are:
>
> - Motion of charged particles in transverse magnetic fields (radius, time period, helical pitch).
> - Biot-Savart Law applications (circular coils, straight conductors, axial field ratios).
> - Ampere's Circuital Law (thick cylindrical conductors, infinite sheets, solenoids).
> - Magnetic force between parallel wires and torque on planar loops ($\boldsymbol{\tau} = \mathbf{m} \times \mathbf{B}$).
> - Moving Coil Galvanometer (current and voltage sensitivities, shunts, series multipliers).

2. Foundations: Oersted's Experiment & Vector Conventions

In 1820, Hans Christian Oersted demonstrated that an electric current flowing through a straight conducting wire deflects a nearby magnetic compass needle. The key experimental findings were:

- The compass needle aligns itself **tangentially** to an imaginary circle centered on the wire and lying in a plane perpendicular to the wire.
- Reversing the direction of current completely reverses the needle orientation.
- Deflection increases when the current I is increased or when the compass needle is brought closer to the wire.
- Iron filings sprinkled on a plane normal to the wire align in **concentric circles** centered on the conductor.

BOSONANCE



Tutorial Diagram 1: Magnetic field lines and tangential needle deflections for currents perpendicular to the plane of paper. Colored arrowheads represent North poles.

Standard Plane Conventions

In electromagnetism, three-dimensional geometry is mapped to two dimensions using arrow-based conventions:

- **Dot $(\odot)$:** Points directed normally outwards from the plane toward the viewer (like the sharp tip of an approaching arrow).
- **Cross $(\otimes)$:** Points directed normally inwards into the plane away from the viewer (like the cross-feathers of a departing arrow).

3. The Lorentz Force on a Moving Charge

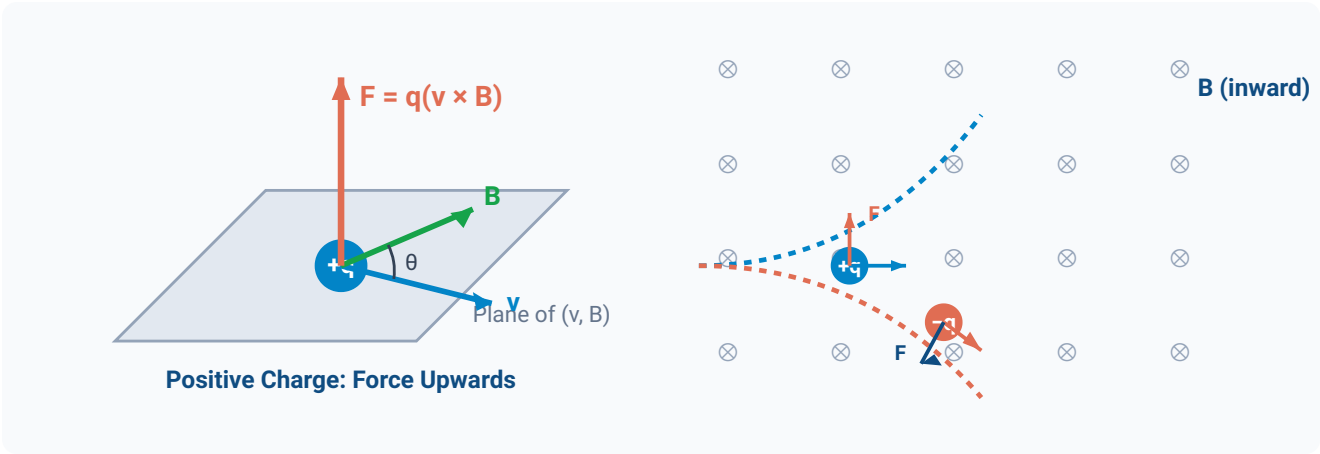
When a test charge q moves with instantaneous velocity $\mathbf{v}$ in a region permeated by both an electric field $\mathbf{E}(\mathbf{r})$ and a magnetic field $\mathbf{B}(\mathbf{r})$, the total mechanical force exerted on the charge is the **Lorentz Force**:

$$\mathbf{F} = \mathbf{F}_{\text{electric}} + \mathbf{F}_{\text{magnetic}} = q\mathbf{E} + q(\mathbf{v} \times \mathbf{B})$$

Critical Properties of the Magnetic Force $\mathbf{F}_m = q(\mathbf{v} \times \mathbf{B})$

1. **Dependence on Charge and Sign:** The magnitude is proportional to $|q|$, and the vector direction for a negative charge (e.g., electron) is strictly opposite to that for a positive charge (e.g., proton or alpha particle).
2. **Cross-Product Geometry:** $\mathbf{F}_m = qvB \sin\theta\, \hat{\mathbf{n}}$, where θ is the angle between $\mathbf{v}$ and $\mathbf{B}$.
   - If $\theta = 0^\circ$ or 180° (motion parallel or anti-parallel to $\mathbf{B}$), $\sin\theta = 0 \implies \mathbf{F}_m = 0$.
   - If $\theta = 90^\circ$ (motion perpendicular to $\mathbf{B}$), the magnetic force attains its maximum magnitude: $F_{m,\max} = qvB$.
3. **Zero Force on Stationary Charges:** If $v = 0$, $\mathbf{F}_m = 0$. Static electric charges neither produce steady magnetic fields nor feel forces from static magnetic fields.
4. **Perpendicularity & Zero Work Done:** Because $\mathbf{F}_m$ is perpendicular to $\mathbf{v}$ at every instant:
   $$\text{Power } P = \mathbf{F}_m \cdot \mathbf{v} = q(\mathbf{v} \times \mathbf{B}) \cdot \mathbf{v} \equiv 0$$
   The magnetic force does **strictly zero mechanical work** on a free charged particle. Consequently, a static magnetic field cannot alter the kinetic energy or speed of a charged particle; it only deflects the direction of momentum.

BOSONANCE



Tutorial Diagram 2: (Left) Cross-product geometry of magnetic Lorentz force. (Right) Opposite deflections of positive and negative charges traversing a transverse magnetic field.

SI Unit and Dimensions of Magnetic Field

From the relation $F = qvB\sin\theta$, we define the magnitude of B as:

$$B = \frac{F}{qv\sin\theta}$$

The SI unit of magnetic field is the **tesla (T)**, named in honor of Nikola Tesla:

$$1\,\mathrm{T} = 1\,\frac{\mathrm{N}}{\mathrm{C}\cdot\mathrm{m/s}} = 1\,\frac{\mathrm{N}}{\mathrm{A}\cdot\mathrm{m}} = 1\,\mathrm{kg}\cdot\mathrm{s}^{-2}\cdot\mathrm{A}^{-1}$$

Dimensional Formula of $\mathbf{B}$:

$$[B] = \frac{[F]}{[q][v]} = \frac{[\mathrm{MLT}^{-2}]}{[\mathrm{AT}][\mathrm{LT}^{-1}]} = [\mathrm{ML}^{0}\mathrm{T}^{-2}\mathrm{A}^{-1}] = [\mathrm{MT}^{-2}\mathrm{A}^{-1}]$$

A common non-SI unit is the **gauss (G)**:

$$1\,\text{gauss} = 10^{-4}\,\text{tesla} \quad (1\,\mathrm{T} = 10^{4}\,\mathrm{G})$$

The Earth's magnetic field at the surface is typically of the order of $0.36\,\mathrm{G} \approx 3.6\times10^{-5}\,\mathrm{T}$.

4. Magnetic Force on a Current-Carrying Conductor

An electric current in a solid conductor consists of a large collection of conduction electrons drifting with an average drift velocity $\mathbf{v}_d$. If the conductor has uniform cross-sectional area A, length l, and number density of mobile charge carriers n, the total number of carriers is $N = nAl$.

The net magnetic force on these mobile electrons (each bearing charge $q = -e$) is:

$$\mathbf{F} = (nAl)\,q\,(\mathbf{v}_d \times \mathbf{B})$$

Since current density is $\mathbf{j} = nq\mathbf{v}_d$ and total current across the section is $I = |\mathbf{j}|A$, we obtain:

$$\mathbf{F} = (\mathbf{j}Al) \times \mathbf{B} = I(\mathbf{l} \times \mathbf{B})$$

$$\mathbf{F} = I(\mathbf{l} \times \mathbf{B})$$

where $\mathbf{l}$ is a vector having magnitude equal to the length l and pointing in the direction of conventional electric current I.

For a wire of arbitrary curved geometry in a non-uniform or uniform field, we divide the conductor into infinitesimal elements $d\mathbf{l}$ and integrate:

$$\mathbf{F} = \int I\,(d\mathbf{l} \times \mathbf{B})$$

BOSONANCE

In a *uniform* magnetic field $\mathbf{B}$, this integral simplifies remarkably:

$$\mathbf{F} = I\left(\int d\mathbf{l}\right) \times \mathbf{B} = I(\mathbf{L}' \times \mathbf{B})$$

where $\mathbf{L}'$ is the straight-line displacement vector connecting the starting point directly to the end point of the curved wire. For any closed current loop in a uniform $\mathbf{B}$, $\oint d\mathbf{l} = 0 \implies \mathbf{F}_{\text{net}} = 0$.

WORKED EXAMPLE 1 • MECHANICAL EQUILIBRIUM IN MAGNETIC FIELD

Problem: A straight horizontal copper wire of length $1.5\,\text{m}$ and mass $200\,\text{g}$ carries a steady current of $2.0\,\text{A}$. It is suspended stationary in mid-air by applying a uniform horizontal magnetic field $\mathbf{B}$. Determine the minimum magnitude and direction of this magnetic field. (Take $g = 9.8\,\text{m/s}^2$).

Solution:

For mid-air suspension without physical supports, the upward magnetic force $\mathbf{F}_m$ must exactly balance the downward gravitational weight:

$$F_m = mg \implies IlB\sin\theta = mg$$

To achieve the minimum field strength B, the field must be perpendicular to the wire ($\theta = 90^\circ \implies \sin\theta = 1$):

$$B = \frac{mg}{Il}$$

Substituting the given values:

$$m = 200\,\text{g} = 0.20\,\text{kg}, \quad l = 1.5\,\text{m}, \quad I = 2.0\,\text{A}$$

$$B = \frac{0.20 \times 9.8}{2.0 \times 1.5} = \frac{1.96}{3.0} \approx 0.653\,\text{T}$$

Direction: If current flows along $+\hat{\mathbf{i}}$ (East), then for $\mathbf{F}_m = I(\mathbf{l} \times \mathbf{B})$ to be directed vertically upward ($+\hat{\mathbf{k}}$), $\hat{\mathbf{i}} \times \hat{\mathbf{B}} = +\hat{\mathbf{k}}$. Hence, $\mathbf{B}$ must point horizontally into the page along $-\hat{\mathbf{j}}$ (North).

5. Dynamics of Charged Particles in a Uniform Magnetic Field

Consider a particle of mass m and charge q injected into a uniform magnetic field $\mathbf{B}$ with initial velocity $\mathbf{v}$. We resolve $\mathbf{v}$ into two components:

$$\mathbf{v} = \mathbf{v}_{\parallel} + \mathbf{v}_{\perp}$$

where $\mathbf{v}_{\parallel}$ is parallel to $\mathbf{B}$, and $\mathbf{v}_{\perp}$ is perpendicular to $\mathbf{B}$.

Case 1: Transverse Injection ($\mathbf{v} \perp \mathbf{B}$, $\theta = 90^\circ$)

Here $\mathbf{v}_{\parallel} = 0$. The magnetic force is constantly perpendicular to the velocity:

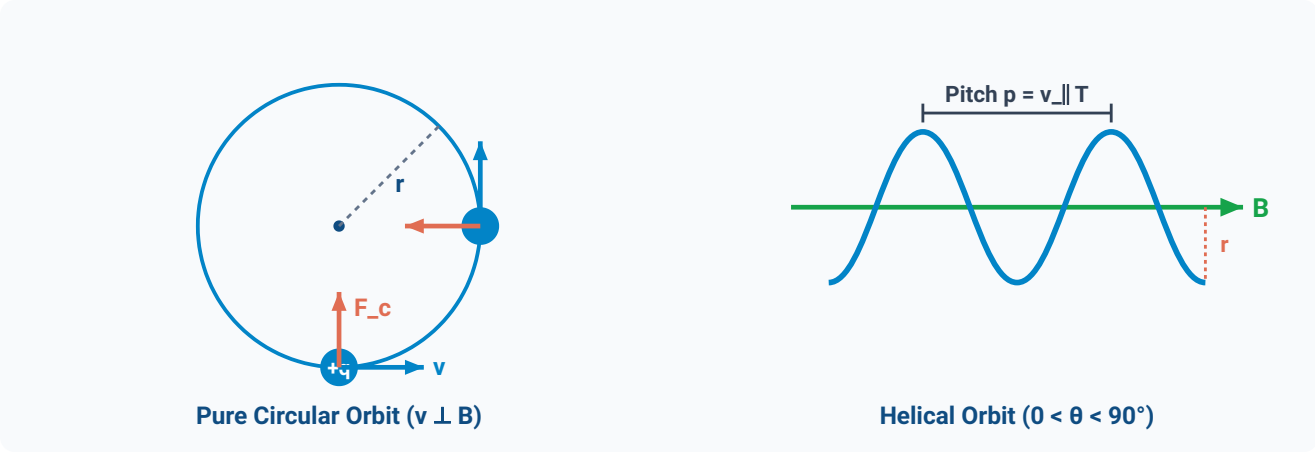
$$F_m = qvB$$

This sideways force provides the necessary centripetal acceleration to bend the particle into a uniform circular orbit:

$$\frac{mv^2}{r} = qvB \implies r = \frac{mv}{qB} = \frac{p}{qB} = \frac{\sqrt{2mK}}{qB} = \frac{\sqrt{2mqV_0}}{qB}$$

where p is linear momentum, K is kinetic energy, and V_0 is the accelerating potential difference ($K = qV_0$).

BOSONANCE



Tutorial Diagram 3: Circular motion under purely transverse magnetic field and helical progression when velocity has a longitudinal component.

Orbital Period, Cyclotron Frequency, and Angular Velocity

The angular velocity ω and time period T for one full circular revolution are:

$$\omega = \frac{v}{r} = \frac{qB}{m}$$

$$T = \frac{2\pi r}{v} = \frac{2\pi}{\omega} = \frac{2\pi m}{qB}$$

The orbital revolution frequency (known historically as the **cyclotron frequency** ν_c) is:

$$\nu_c = \frac{1}{T} = \frac{qB}{2\pi m}$$

> **Key Invariance Law for NEET UG**
>
> Both time period T and cyclotron frequency ν_c are **strictly independent of the particle's speed v, orbital radius r, and kinetic energy K**. Faster particles simply describe wider circles in the exact same duration of time. This resonance principle enables synchronized radiofrequency acceleration in particle accelerators.

Case 2: Oblique Injection — Helical Trajectory

When the velocity vector makes an arbitrary angle θ with $\mathbf{B}$:

$$v_{\parallel} = v\cos\theta, \quad v_{\perp} = v\sin\theta$$

The parallel component $v_{\parallel}$ experiences zero magnetic force ($F_{\parallel} = qv_{\parallel}B\sin 0^\circ = 0$), leaving the longitudinal velocity constant. The transverse component $v_{\perp}$ undergoes circular motion with radius:

$$r_{\text{helix}} = \frac{mv_{\perp}}{qB} = \frac{mv\sin\theta}{qB}$$

The linear distance traveled parallel to the magnetic field in one revolution is termed the **pitch (p)** of the helix:

$$p = v_{\parallel}T = (v\cos\theta)\left(\frac{2\pi m}{qB}\right) = \frac{2\pi mv\cos\theta}{qB}$$

BOSONANCE

WORKED EXAMPLE 2 • ELECTRON ORBIT IN MAGNETIC CHAMBER

Problem: In an experimental vacuum chamber, an electron ($m = 9.0 \times 10^{-31}\,\text{kg}$, $e = 1.6 \times 10^{-19}\,\text{C}$) is projected with speed $v = 3.0 \times 10^{7}\,\text{m/s}$ perpendicular to a uniform magnetic field $B = 6.0 \times 10^{-4}\,\text{T}$. Calculate: (a) orbital radius r, (b) orbital revolution frequency ν, and (c) kinetic energy in keV.

Solution:

(a) Radius of circular path:

$$r = \frac{mv}{eB} = \frac{(9.0 \times 10^{-31}\,\text{kg}) \times (3.0 \times 10^{7}\,\text{m/s})}{(1.6 \times 10^{-19}\,\text{C}) \times (6.0 \times 10^{-4}\,\text{T})} = \frac{27.0 \times 10^{-24}}{9.6 \times 10^{-23}} = 0.281\,\text{m} \approx 28\,\text{cm}$$

(b) Frequency of revolution:

$$\nu = \frac{eB}{2\pi m} = \frac{(1.6 \times 10^{-19}) \times (6.0 \times 10^{-4})}{2 \times 3.1416 \times (9.0 \times 10^{-31})} = \frac{9.6 \times 10^{-23}}{5.655 \times 10^{-30}} \approx 1.7 \times 10^{7}\,\text{Hz} = 17\,\text{MHz}$$

(c) Kinetic energy in keV:

$$K = \frac{1}{2}mv^2 = \frac{1}{2}(9.0 \times 10^{-31}\,\text{kg}) \times (3.0 \times 10^{7}\,\text{m/s})^2 = 4.05 \times 10^{-16}\,\text{J}$$

Converting to electron-volts ($1\,\text{eV} = 1.6 \times 10^{-19}\,\text{J}$):

$$K = \frac{4.05 \times 10^{-16}\,\text{J}}{1.6 \times 10^{-19}\,\text{J/eV}} = 2531\,\text{eV} \approx 2.53\,\text{keV}$$

6. The Biot-Savart Law

The **Biot-Savart Law** is the fundamental law of magnetostatics relating a steady current element $Id\mathbf{l}$ to the infinitesimal magnetic field $d\mathbf{B}$ it produces at an arbitrary point P located at displacement vector $\mathbf{r}$.

$$d\mathbf{B} = \frac{\mu_0}{4\pi}\frac{I(d\mathbf{l} \times \mathbf{r})}{r^3} = \frac{\mu_0}{4\pi}\frac{I(d\mathbf{l} \times \hat{\mathbf{r}})}{r^2}$$

The magnitude is:

$$|d\mathbf{B}| = \frac{\mu_0}{4\pi}\frac{Idl\sin\theta}{r^2}$$

where θ is the angle between the current element vector $d\mathbf{l}$ and the position vector $\mathbf{r}$.



BOSONANCE

Tutorial Diagram 4: Geometric representation of the Biot-Savart Law. The field dB is perpendicular to the plane defined by dl and r.

Physical Constants & Relativistic Connection

The constant μ_0 is the **permeability of free space**:

$$\frac{\mu_0}{4\pi} = 10^{-7}\,\mathrm{T\cdot m/A} \implies \mu_0 = 4\pi \times 10^{-7}\,\mathrm{T\cdot m\cdot A^{-1}}$$

Combining μ_0 with the permittivity of free space $\varepsilon_0 = \frac{1}{4\pi\times 9\times 10^9}\,\mathrm{C^2/(N\cdot m^2)}$:

$$\mu_0\varepsilon_0 = \left(\frac{\mu_0}{4\pi}\right)(4\pi\varepsilon_0) = 10^{-7} \times \frac{1}{9\times 10^9} = \frac{1}{(3\times 10^8)^2} = \frac{1}{c^2}$$

This profound relationship shows that the speed of light in vacuum is:

$$c = \frac{1}{\sqrt{\mu_0\varepsilon_0}}$$

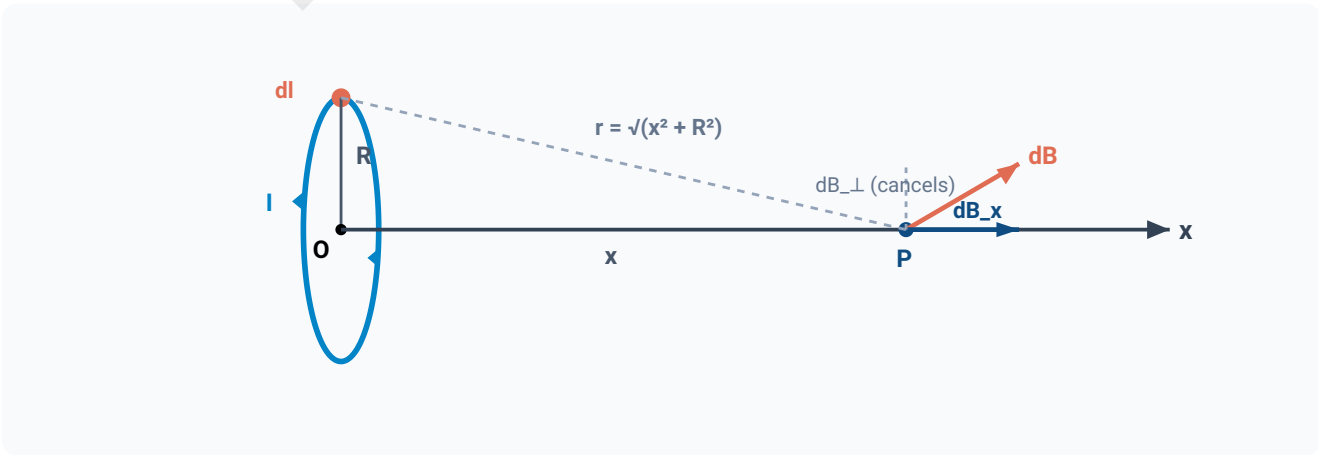
Comparison: Biot-Savart Law vs. Coulomb's Law

Property	Electrostatic Field (Coulomb's Law)	Magnetic Field (Biot-Savart Law)
Source	Scalar source: Electric charge dq.	Vector source: Current element $Id\mathbf{l}$.
Distance Dependence	Inverse-square law ($E \propto 1/r^2$); long-range.	Inverse-square law ($dB \propto 1/r^2$); long-range.
Directionality	Radially along the displacement vector $\hat{\mathbf{r}}$.	Perpendicular to both $d\mathbf{l}$ and $\hat{\mathbf{r}}$ (plane normal).
Angular Dependence	Spherically symmetric; no angle dependence.	Depends on $\sin\theta$; zero along the wire axis ($\theta = 0^\circ, 180^\circ$).
Superposition	Linear superposition applies ($\mathbf{E} = \sum \mathbf{E}_i$).	Linear superposition applies ($\mathbf{B} = \sum \mathbf{B}_i$).

7. Magnetic Field Due to Circular Loops and Coils

Magnetic Field on the Axis of a Circular Current Loop

Consider a circular loop of radius R lying in the y-z plane, carrying steady current I. We wish to determine the field at point P along the axis (x-axis) at distance x from the center O.



Tutorial Diagram 5: Axial magnetic field of a circular current loop. Perpendicular components cancel in pairs, leaving only axial components directed along +x.

BOSONANCE

For every element $d\mathbf{l}$, the displacement vector $\mathbf{r}$ to the axial point has magnitude $r = \sqrt{x^2 + R^2}$. Because $d\mathbf{l}$ lies in the y-z plane while $\mathbf{r}$ is inclined, they are mutually perpendicular: $|d\mathbf{l} \times \mathbf{r}| = r\, dl$.

$$dB = \frac{\mu_0}{4\pi} \frac{Idl}{x^2 + R^2}$$

By axial symmetry, the perpendicular components $dB_\perp = dB \sin\theta$ cancel when integrated over diametrically opposite elements. Only the axial components $dB_x = dB \cos\theta$ survive:

$$\cos\theta = \frac{R}{\sqrt{x^2 + R^2}}$$

$$dB_x = \frac{\mu_0 Idl}{4\pi(x^2 + R^2)} \frac{R}{(x^2 + R^2)^{1/2}} = \frac{\mu_0 IR}{4\pi(x^2 + R^2)^{3/2}} dl$$

Integrating dl over the entire circular circumference ($\oint dl = 2\pi R$):

$$B_x = \frac{\mu_0 IR^2}{2(x^2 + R^2)^{3/2}}$$

For an N-turn closely wound coil:

$$B_{\text{axial}} = \frac{\mu_0 NIR^2}{2(x^2 + R^2)^{3/2}}$$

Special Cases for Circular Geometries

1. **At the Center of the Loop ($x = 0$):**
$$B_{\text{center}} = \frac{\mu_0 IR^2}{2(R^2)^{3/2}} = \frac{\mu_0 I}{2R} \quad \left(\text{For } N \text{ turns: } B_0 = \frac{\mu_0 NI}{2R}\right)$$
2. **At Large Axial Distances ($x \gg R$):**
$$B \approx \frac{\mu_0 IR^2}{2x^3} = \frac{\mu_0 (I \cdot \pi R^2)}{2\pi x^3} = \frac{\mu_0 m}{2\pi x^3} = \frac{\mu_0}{4\pi} \frac{2m}{x^3}$$
where $m = IA = I(\pi R^2)$ is the magnetic dipole moment of the circular loop.
3. **Field at Distance $x = R$:**
$$B(x = R) = \frac{\mu_0 IR^2}{2(R^2 + R^2)^{3/2}} = \frac{\mu_0 IR^2}{2(2R^2)^{3/2}} = \frac{\mu_0 I}{4\sqrt{2}R} = \frac{B_0}{2\sqrt{2}} \approx 0.353\, B_0$$
4. **Circular Arc Subtending Angle ϕ at Center:**
$$B_{\text{arc}} = \left(\frac{\phi}{2\pi}\right) \frac{\mu_0 I}{2R}$$
   - **Semicircular Arc ($\phi = \pi$):** $B = \frac{1}{2}\left(\frac{\mu_0 I}{2R}\right) = \frac{\mu_0 I}{4R}$.
   - **Quarter-Circle Arc ($\phi = \pi/2$):** $B = \frac{1}{4}\left(\frac{\mu_0 I}{2R}\right) = \frac{\mu_0 I}{8R}$.

BOSONANCE

WORKED EXAMPLE 3 • FIELD AT CENTER OF TIGHTLY WOUND COIL

Problem: A tightly wound circular coil having $N = 100$ turns and radius $R = 10\,\text{cm}$ carries a steady current of $1.0\,\text{A}$. Calculate the magnitude of the magnetic field at the center of the coil.

Solution:

Given: $N = 100$, $I = 1.0\,\text{A}$, $R = 10\,\text{cm} = 0.10\,\text{m}$. Using the center formula:

$$B = \frac{\mu_0 N I}{2R}$$

$$B = \frac{(4\pi \times 10^{-7}\,\text{T}\cdot\text{m/A}) \times 100 \times 1.0\,\text{A}}{2 \times 0.10\,\text{m}} = \frac{4\pi \times 10^{-5}}{0.20} = 2\pi \times 10^{-4}\,\text{T} \approx 6.28 \times 10^{-4}\,\text{T}$$

8. Ampere's Circuital Law

Ampere's Circuital Law provides an elegant method to calculate magnetic fields for current distributions possessing high spatial symmetry, serving an operational role analogous to Gauss's Law in electrostatics.

$$\oint_C \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{\text{enclosed}}$$

The line integral of the magnetic field $\mathbf{B}$ around any closed imaginary boundary loop C equals μ_0 times the net steady electric current I_{enclosed} crossing the open surface bounded by C.

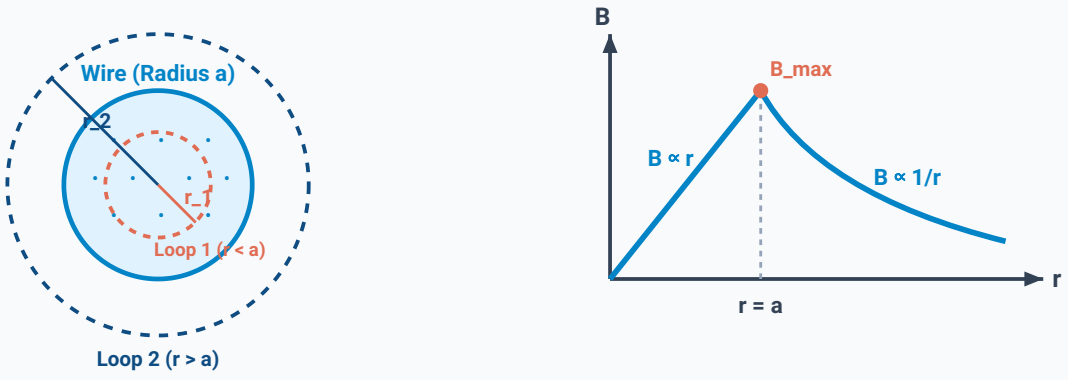
Sign Convention (Right-Hand Rule)

Curl the fingers of the right hand in the direction of traversal along the closed Amperian loop. The extended right-hand thumb points in the direction of currents regarded as **positive**. Currents flowing in the reverse sense are taken as **negative**.

Application 1: Long, Straight Cylindrical Conductor of Radius a

Consider an infinitely long, straight solid cylindrical wire of radius a carrying a uniformly distributed current I across its cross-section. The current density is:

$$j = \frac{I}{\pi a^2}$$



Tutorial Diagram 6: (Left) Concentric Amperian loops inside and outside a solid cylindrical conductor. (Right) Radial variation of magnetic field magnitude B as a function of distance r from the wire axis.

BOSONANCE

Region 1: Exterior to the wire ($r \geq a$)

Choosing a circular Amperian loop of radius r concentric with the wire:

$$\oint \mathbf{B} \cdot d\mathbf{l} = B(2\pi r) = \mu_0 I \implies B_{\text{out}} = \frac{\mu_0 I}{2\pi r} \quad (B \propto r^{-1})$$

Region 2: Interior to the wire ($r < a$)

Choosing a concentric circular loop of radius $r < a$, the fraction of current enclosed is:

$$I_{\text{enclosed}} = j(\pi r^2) = \frac{I}{\pi a^2}(\pi r^2) = I\frac{r^2}{a^2}$$

Applying Ampere's Law:

$$B(2\pi r) = \mu_0 \left(I\frac{r^2}{a^2} \right) \implies B_{\text{in}} = \left(\frac{\mu_0 I}{2\pi a^2} \right) r \quad (B \propto r)$$

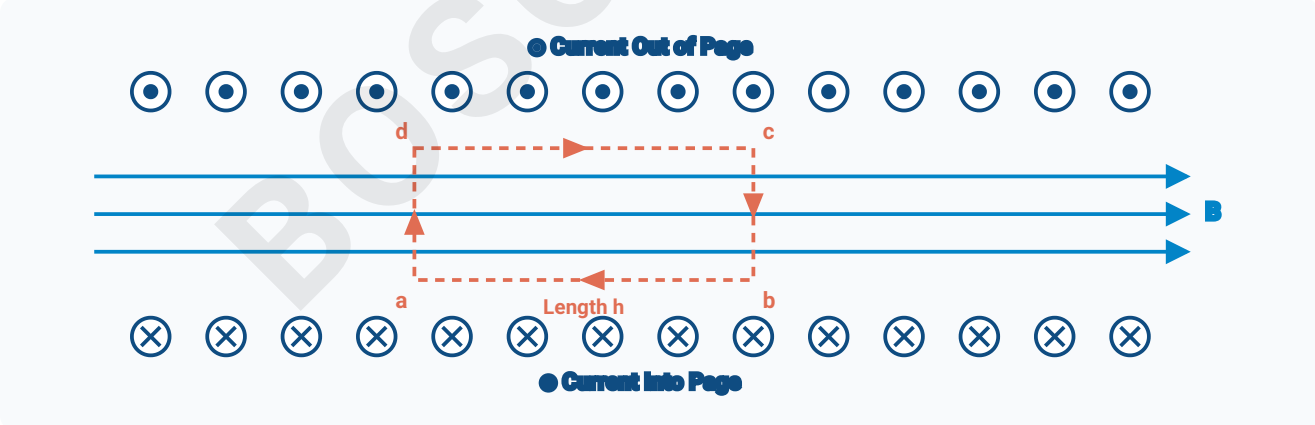
At the axis ($r = 0$), $B = 0$. The field increases linearly to its maximum value at the surface $r = a$:

$$B_{\text{max}} = \frac{\mu_0 I}{2\pi a}$$

9. The Long Solenoid

A **solenoid** is a long helical winding of insulated wire wound tightly around a cylindrical core. For a *long* solenoid (length $L \gg$ radius R):

- Inside the solenoid, the magnetic field is remarkably **uniform, strong, and directed strictly along the central axis**.
- Outside the solenoid, the field lines spread over a large volume and cancel out, rendering the exterior field **vanishingly small** ($B_{\text{out}} \approx 0$).



Tutorial Diagram 7: Magnetic field of an ideal long solenoid and the rectangular Amperian loop abcd used to derive the axial field formula.

Derivation using Ampere's Law

Consider a rectangular Amperian loop $abcd$ of axial length h:

$$\oint_{abcd} \mathbf{B} \cdot d\mathbf{l} = \int_a^b \mathbf{B} \cdot d\mathbf{l} + \int_b^c \mathbf{B} \cdot d\mathbf{l} + \int_c^d \mathbf{B} \cdot d\mathbf{l} + \int_d^a \mathbf{B} \cdot d\mathbf{l}$$

Evaluating each segment:

- Along interior segment ab: $\mathbf{B}$ is uniform and parallel to $d\mathbf{l} \implies \int_a^b \mathbf{B} \cdot d\mathbf{l} = Bh$.

BOSONANCE

- Along transverse segments bc and da: $\mathbf{B}$ is perpendicular to the path $\implies \mathbf{B}\cdot d\mathbf{l} = 0$.
- Along exterior segment cd: $\mathbf{B}_{\text{out}} = 0 \implies \int_c^d \mathbf{B}\cdot d\mathbf{l} = 0$.

Therefore, $\oint_{abcd} \mathbf{B}\cdot d\mathbf{l} = Bh$.

If n is the number of turns per unit length, the total turns threaded through the loop is nh, and enclosed current is $I_{\text{enclosed}} = (nh)I$.

$$Bh = \mu_0(nhI) \implies B = \mu_0 nI$$

$$B_{\text{interior}} = \mu_0 nI = \mu_0 \left(\frac{N}{L}\right) I$$

At the extreme open end of a semi-infinite solenoid:

$$B_{\text{end}} = \frac{1}{2}\mu_0 nI$$

WORKED EXAMPLE 4 • MULTI-LAYERED SOLENOID

Problem: A solenoid of length $80\,\text{cm}$ has 5 concentric layers of windings of 400 turns each. Its diameter is $1.8\,\text{cm}$. If the current carried is $8.0\,\text{A}$, estimate the magnitude of $\mathbf{B}$ near the center of the solenoid on its axis.

Solution:

Total number of turns $N = 5 \times 400 = 2000$ turns.
Length of solenoid $L = 80\,\text{cm} = 0.80\,\text{m}$.
Number of turns per unit length:

$$n = \frac{N}{L} = \frac{2000}{0.80} = 2500\,\text{turns/m}$$

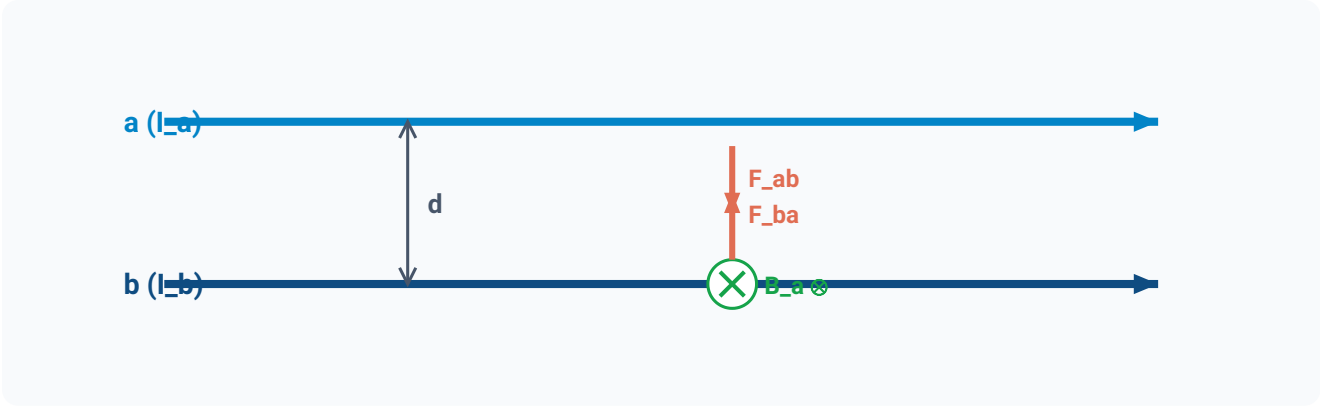
Because $L = 0.80\,\text{m}$ is far larger than diameter $d = 0.018\,\text{m}$ ($L/d \approx 44 \gg 1$), the long solenoid approximation applies:

$$B = \mu_0 nI = (4\pi \times 10^{-7}\,\text{T}\cdot\text{m/A}) \times 2500\,\text{m}^{-1} \times 8.0\,\text{A}$$
$$B = 4\pi \times 10^{-7} \times 20000 = 8\pi \times 10^{-3}\,\text{T} \approx 2.51 \times 10^{-2}\,\text{T} = 25.1\,\text{mT}$$

10. Mutual Force Between Parallel Current-Carrying Conductors

Consider two infinitely long, thin parallel conductors a and b separated by a distance d in vacuum, carrying steady currents I_a and I_b.

BOSONANCE



Tutorial Diagram 8: Magnetic interaction between two parallel conductors carrying currents in the same direction, generating equal and opposite attractive forces.

The magnetic field produced by wire a at any point along wire b has magnitude:

$$B_a = \frac{\mu_0 I_a}{2\pi d}$$

By the right-hand grip rule, $\mathbf{B}_a$ is directed into the plane of the page at wire b. Consequently, a segment of length L of wire b experiences a Lorentz force:

$$\mathbf{F}_{ba} = I_b(\mathbf{L} \times \mathbf{B}_a) \implies F_{ba} = I_b L B_a = \frac{\mu_0 I_a I_b}{2\pi d} L$$

By the vector cross product, $\mathbf{F}_{ba}$ points directly toward wire a. By symmetry, wire a experiences an identical attraction toward wire b:

$$\mathbf{F}_{ab} = -\mathbf{F}_{ba} \quad \text{(Satisfies Newton's Third Law)}$$

> **The Universal Rule of Magnetic Currents**
>
> **Parallel currents (same direction) ATTRACT.**
> **Anti-parallel currents (opposite directions) REPEL.**
> *Contrast with Electrostatics:* Like charges repel, whereas like currents attract!

Force per Unit Length and the SI Definition of the Ampere

The force per unit length $f = F/L$ is given by:

$$f = \frac{\mu_0 I_a I_b}{2\pi d}$$

Setting $I_a = I_b = 1\,\text{A}$ and $d = 1\,\text{m}$ in vacuum:

$$f = \frac{(4\pi \times 10^{-7}) \times 1 \times 1}{2\pi \times 1} = 2 \times 10^{-7}\,\text{N/m}$$

> **Theoretical SI Definition of 1 Ampere (Adopted 1946)**
>
> **The Ampere** is that constant steady current which, if maintained in two straight parallel conductors of infinite length and negligible circular cross-section, placed $1\,\text{metre}$ apart in vacuum, would produce between these conductors a force equal to exactly $2 \times 10^{-7}\,\text{newtons}$ per metre of length.

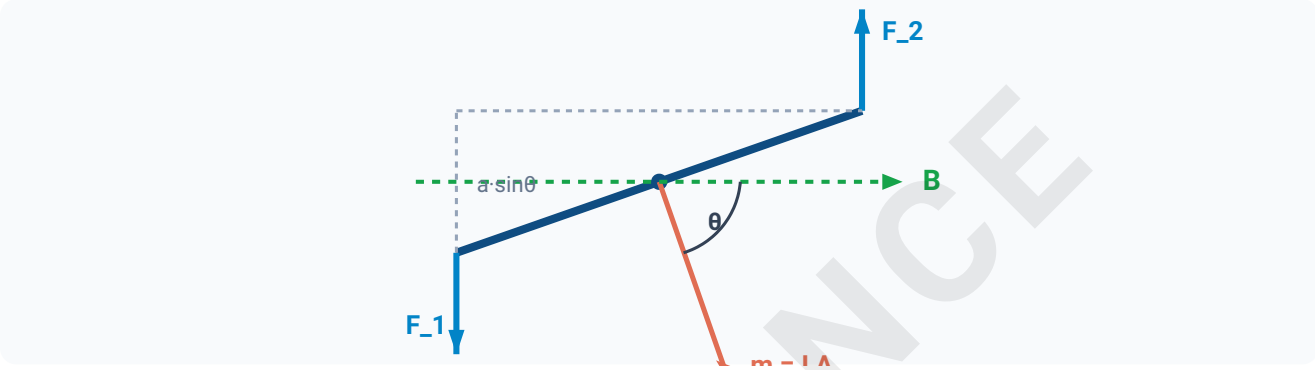
BOSONANCE

11. Torque on a Current Loop & Magnetic Dipole Moment

Torque on a Rectangular Loop in a Uniform Field

Consider a rectangular planar loop $ABCD$ of sides a and b carrying current I in a uniform field $\mathbf{B}$. Let the normal to the plane of the loop (area vector $\mathbf{A}$) make an angle θ with $\mathbf{B}$.

- Forces on the opposite transverse sides BC and DA are equal in magnitude ($F = IaB\sin(90^\circ - \theta)$), collinear along the loop axis, and opposite in direction. They cancel completely with zero net force and zero net torque.
- Forces on sides AB and CD each have magnitude $F_1 = F_2 = IbB$. They are equal and opposite, so $\mathbf{F}_{\text{net}} = 0$. However, they are **non-collinear**, separated by a perpendicular lever arm $a\sin\theta$.



Tutorial Diagram 9: Top end-on view of a current loop showing the couple of forces producing torque $\boldsymbol{\tau} = \mathbf{m} \times \mathbf{B}$.

The magnitude of this deflecting torque is:

$$\tau = F_1\left(\frac{a}{2}\sin\theta\right) + F_2\left(\frac{a}{2}\sin\theta\right) = (IbB)(a\sin\theta) = I(ab)B\sin\theta = IAB\sin\theta$$

Defining the **magnetic dipole moment vector** of the loop as:

$$\mathbf{m} = NI\mathbf{A} \quad (\text{SI Unit: A} \cdot \text{m}^2 \text{ or J/T})$$

the torque vector becomes:

$$\boldsymbol{\tau} = \mathbf{m} \times \mathbf{B}$$

Equilibrium States and Potential Energy

- **Stable Equilibrium ($\theta = 0^\circ$):** $\mathbf{m}$ is aligned parallel to $\mathbf{B}$. $\tau = 0$, potential energy $U = -mB$ is at a minimum. Any small angular perturbation generates a restoring torque directed back toward alignment.
- **Unstable Equilibrium ($\theta = 180^\circ$):** $\mathbf{m}$ is antiparallel to $\mathbf{B}$. $\tau = 0$, potential energy $U = +mB$ is maximum. Any slight displacement produces a torque accelerating the loop away from this state.

BOSONANCE

WORKED EXAMPLE 5 • ROTATIONAL DYNAMICS OF A MAGNETIC COIL

Problem: A circular coil of 100 turns and radius $10\,\text{cm}$ carries a current of $3.2\,\text{A}$. The coil is suspended in a uniform horizontal field $B = 2.0\,\text{T}$ such that initially its magnetic moment $\mathbf{m}$ is parallel to $\mathbf{B}$. If the coil rotates by 90° under the influence of the field, calculate: (a) initial and final torques, and (b) the angular speed acquired by the coil given its moment of inertia about the rotation axis is $J = 0.10\,\text{kg}\cdot\text{m}^2$.

Solution:

(a) Magnetic moment & Torques:

$$m = NI(\pi R^2) = 100 \times 3.2\,\text{A} \times (3.1416 \times 0.10^2\,\text{m}^2) \approx 10.0\,\text{A}\cdot\text{m}^2$$

Initial position ($\theta = 0^\circ$): $\tau_i = mB\sin 0^\circ = 0$.
Final position ($\theta = 90^\circ$): $\tau_f = mB\sin 90^\circ = 10.0 \times 2.0 = 20\,\text{N}\cdot\text{m}$.

(b) Angular speed ω_f: From the work-energy theorem for rotational motion:

$$W = \Delta K_{\text{rot}} \implies \int_0^{\pi/2} \tau\, d\theta = \frac{1}{2}J\omega_f^2$$

$$\int_0^{\pi/2} mB\sin\theta\, d\theta = mB[-\cos\theta]_0^{\pi/2} = mB(0-(-1)) = mB$$

Equating to rotational kinetic energy:

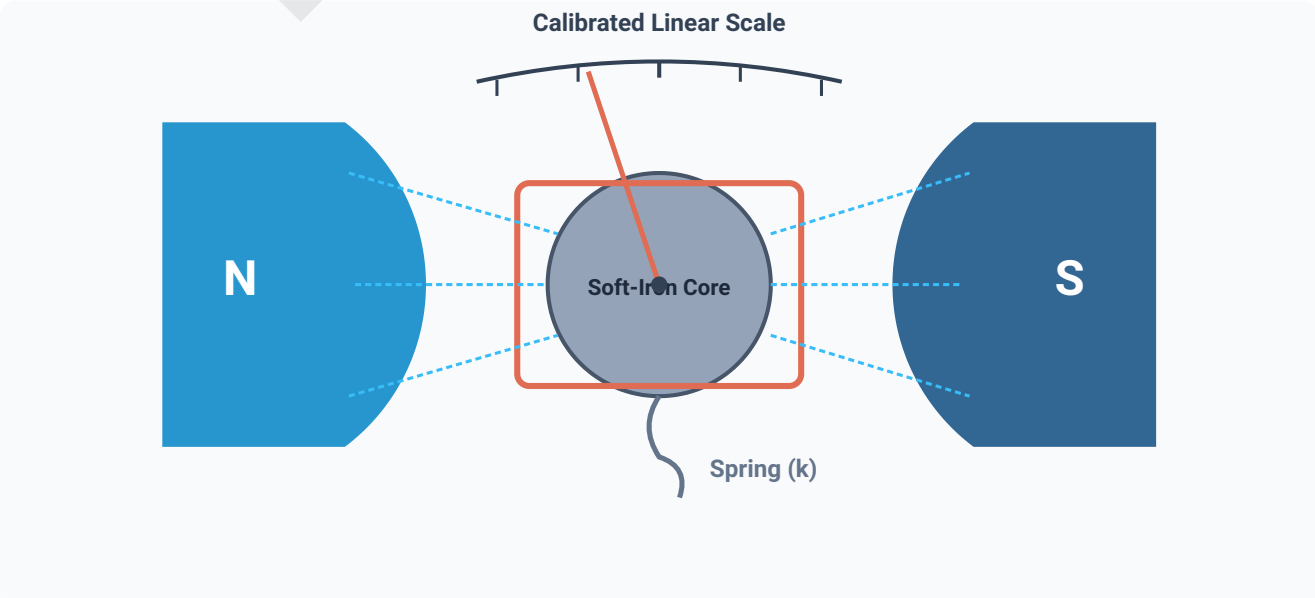
$$\frac{1}{2}J\omega_f^2 = mB \implies \omega_f = \sqrt{\frac{2mB}{J}}$$

Substituting numerical parameters:

$$\omega_f = \sqrt{\frac{2 \times (10.0 \times 2.0)}{0.10}} = \sqrt{\frac{40}{0.10}} = \sqrt{400} = 20\,\text{rad/s}$$

12. The Moving Coil Galvanometer (MCG)

The **Moving Coil Galvanometer** is the premier electromagnetic instrument used for the detection and sensitive measurement of tiny electric currents.



BOSONANCE

Tutorial Diagram 10: Construction of the Moving Coil Galvanometer featuring concave magnetic pole shoes and a central cylindrical soft iron core providing a uniform radial field.

Key Constructional Features

1. **Cylindrical Soft Iron Core:** Possesses high magnetic permeability; concentrates magnetic flux lines and ensures that the magnetic field is directed **radially** at all angles of rotation.
2. **Concave Magnetic Poles:** Ensures that the plane of the coil remains parallel to the magnetic field vectors at every orientation ($\theta = 90^\circ$), making the deflecting torque strictly independent of coil rotation angle ϕ:
$$\tau_{\text{def}} = NIAB \sin 90^\circ = NIAB$$
3. **Phosphor-Bronze Suspension / Hairspring:** Exerts a restoring elastic torque directly proportional to the twist deflection ϕ:
$$\tau_{\text{restoring}} = k\phi$$
where k is the torsional spring constant (restoring torque per unit radian of twist).

Governing Deflection Formula & Sensitivity

In mechanical rotational equilibrium:

$$\tau_{\text{def}} = \tau_{\text{restoring}} \implies NIAB = k\phi \implies \phi = \left(\frac{NAB}{k}\right) I$$

Because $\frac{NAB}{k}$ is a device constant, the deflection ϕ is **directly proportional to current** I, yielding a perfectly linear scale.

Quantity	Formula	Definition & Units
Current Sensitivity (I_s)	$I_s = \frac{\phi}{I} = \frac{NAB}{k}$	Deflection produced per unit electric current. Measured in rad/A or div/A.
Voltage Sensitivity (V_s)	$V_s = \frac{\phi}{V} = \frac{\phi}{IR_G} = \frac{NAB}{kR_G}$	Deflection produced per unit applied voltage. Measured in rad/V or div/V.

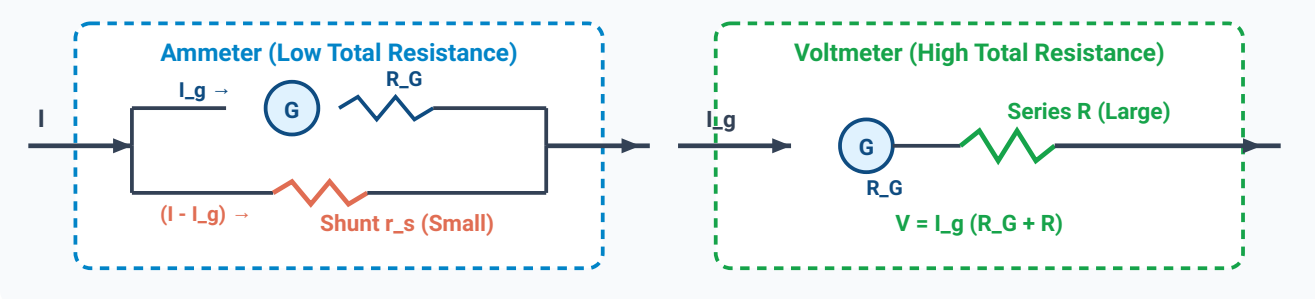
NEET High-Yield Conceptual Trap: Doubling N

If the number of turns N is doubled ($N \to 2N$):
• **Current sensitivity doubles:** $I_s \propto N \implies I_s' = 2I_s$.
• **Voltage sensitivity remains UNCHANGED:** Doubling the number of turns doubles the total length of wire used, which doubles coil resistance ($R_G \to 2R_G$). Hence:

$$V_s' = \frac{(2N)AB}{k(2R_G)} = \frac{NAB}{kR_G} = V_s$$

Conclusion: Increasing current sensitivity does NOT automatically increase voltage sensitivity.

Conversion of Galvanometer into Ammeter and Voltmeter



BOSONANCE

Tutorial Diagram 11: Electrical schematics for converting a moving coil galvanometer into an ammeter (parallel low-resistance shunt) and a voltmeter (series high-resistance multiplier).

Conversion 1: To an Ammeter (Parallel Shunt r_s)

An ammeter must be inserted in series with the load; hence its internal resistance must be negligible so that it does not reduce the circuit current. A **small shunt resistance r_s is connected in parallel** with the galvanometer coil (R_G).
Because the potential drop across the parallel arms is identical:

$$V_{\text{branch}} = I_g R_G = (I - I_g) r_s \implies r_s = \frac{I_g R_G}{I - I_g}$$

Total equivalent resistance of the resulting ammeter:

$$R_A = \frac{R_G r_s}{R_G + r_s} \approx r_s \quad (\text{Since } r_s \ll R_G)$$

An **ideal ammeter** has zero internal resistance ($R_A = 0$).

Conversion 2: To a Voltmeter (Series Multiplier R)

A voltmeter must be connected in parallel across circuit components to measure potential difference; hence it must draw negligible current to prevent loading error. A **large multiplier resistance R is connected in series** with the galvanometer.

$$V = I_g(R_G + R) \implies R = \frac{V}{I_g} - R_G$$

Total resistance of the voltmeter:

$$R_V = R_G + R \approx R \quad (\text{Since } R \gg R_G)$$

An **ideal voltmeter** has infinite internal resistance ($R_V \to \infty$).

13. Master Revision Table & High-Yield Formula Sheet

Physical Quantity / Concept	Standard Mathematical Formulation	Key Variables & Critical Conditions
Lorentz Force	$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$	$\mathbf{F}_m = qvB\sin\theta$; zero work done by magnetic force.
Force on Straight Conductor	$\mathbf{F} = I(\mathbf{l} \times \mathbf{B})$	$\mathbf{l}$ directed along current; $\mathbf{F} = 0$ for closed planar loop in uniform $\mathbf{B}$.
Circular Orbit Radius	$r = \frac{mv}{qB} = \frac{p}{qB} = \frac{\sqrt{2mK}}{qB}$	Valid for $\mathbf{v} \perp \mathbf{B}$. $K = qV_0$.
Cyclotron Period & Frequency	$T = \frac{2\pi m}{qB}, \quad \nu = \frac{qB}{2\pi m}$	**Independent** of velocity v, radius r, and kinetic energy K.
Helical Pitch	$p = v_{\parallel} T = \frac{2\pi m v \cos\theta}{qB}$	Radius of helix is $r = \frac{mv\sin\theta}{qB}$.
Biot-Savart Law	$d\mathbf{B} = \frac{\mu_0}{4\pi}\frac{I(d\mathbf{l}\times\mathbf{r})}{r^3}$	$\frac{\mu_0}{4\pi} = 10^{-7}\,\text{T}\cdot\text{m/A}$; $\mu_0\varepsilon_0 = 1/c^2$.
Circular Loop (Axis)	$B = \frac{\mu_0 I R^2}{2(x^2+R^2)^{3/2}}$	Center ($x=0$): $B_0 = \frac{\mu_0 I}{2R}$. Large distance ($x \gg R$): $B = \frac{\mu_0 m}{2\pi x^3}$.
Ampere's Circuital Law	$\oint \mathbf{B}\cdot d\mathbf{l} = \mu_0 I_{\text{enclosed}}$	Long wire: $B = \frac{\mu_0 I}{2\pi r}$. Thick wire inside ($r < a$): $B \propto r$.
Long Solenoid	$B = \mu_0 n I = \mu_0 \frac{N}{L} I$	Center: $B = \mu_0 n I$; End of semi-infinite solenoid: $B_{\text{end}} = \frac{1}{2}\mu_0 n I$.
Parallel Currents Force	$f = \frac{F}{L} = \frac{\mu_0 I_1 I_2}{2\pi d}$	Parallel attract; anti-parallel repel. 1 Ampere produces $2\times10^{-7}\,\text{N/m}$.

BOSONANCE

Physical Quantity / Concept	Standard Mathematical Formulation	Key Variables & Critical Conditions
Torque on Planar Loop	$\boldsymbol{\tau} = \mathbf{m} \times \mathbf{B} = NIAB \sin\theta$	Stable at $\theta = 0^\circ$ ($U = -mB$); Unstable at $\theta = 180^\circ$ ($U = +mB$).
MCG Deflection & Shunts	$\phi = \left(\frac{NAB}{k}\right) I, \quad r_s = \frac{I_g R_G}{I - I_g}$	Series multiplier for voltmeter: $R = \frac{V}{I_g} - R_G$.

14. NEET UG Exam-Aligned Practice Questions

The following original practice questions match the precise format, style, and cognitive difficulty of the NEET UG examination (including single-correct MCQs, Assertion-Reason pairs, and multi-statement evaluations).

Practice Question 1

NEET Single Correct • Particle Kinematics

A proton, a deuteron ($^2_1\mathrm{H}^+$), and an α-particle ($^4_2\mathrm{He}^{2+}$) having identical kinetic energies enter a uniform magnetic field $\mathbf{B}$ perpendicularly. If r_p, r_d, and r_α represent the radii of their respective circular trajectories, which of the following relations is correct?

(A) $r_p = r_d = r_\alpha$

(B) $r_\alpha = r_p < r_d$

(C) $r_p < r_d < r_\alpha$

(D) $r_d < r_p = r_\alpha$

Correct Answer: (B) $r_\alpha = r_p < r_d$

Detailed Reasoning:
The orbital radius of a charged particle in terms of kinetic energy K is:

$$r = \frac{p}{qB} = \frac{\sqrt{2mK}}{qB}$$

Since K and B are identical for all three particles, $r \propto \frac{\sqrt{m}}{q}$.

- **Proton (p):** $m_p = m, q_p = e \implies r_p \propto \frac{\sqrt{m}}{e} = 1.$
- **Deuteron (d):** $m_d = 2m, q_d = e \implies r_d \propto \frac{\sqrt{2m}}{e} = \sqrt{2} \approx 1.414.$
- **Alpha particle (α):** $m_\alpha = 4m, q_\alpha = 2e \implies r_\alpha \propto \frac{\sqrt{4m}}{2e} = \frac{2\sqrt{m}}{2e} = 1.$

Therefore:

$$r_p = r_\alpha = \frac{r_d}{\sqrt{2}} \implies r_\alpha = r_p < r_d$$

BOSONANCE

Practice Question 2

NEET Single Correct • Biot-Savart Law

A wire carrying steady current I is first formed into a single circular turn of radius R. The magnetic field at its center is B_1. The same wire is subsequently rewound into a closely wound circular coil of 4 turns of radius R'. If the same current I is maintained, the magnetic field at the center of the new coil is B_2. The ratio B_2/B_1 is:

(A) 4

(B) 8

(C) 16

(D) 1

Correct Answer: (C) 16

Detailed Reasoning:
Let L be the total fixed length of the wire.
For the single turn ($N_1 = 1$):

$$L = 2\pi R \implies R = \frac{L}{2\pi}, \quad B_1 = \frac{\mu_0 (1) I}{2R} = \frac{\mu_0 I \pi}{L}$$

When rewound into $N_2 = 4$ turns:

$$L = N_2(2\pi R') = 4(2\pi R') \implies R' = \frac{R}{4}$$

The field at the center of an N-turn coil is:

$$B_2 = \frac{\mu_0 N_2 I}{2R'} = \frac{\mu_0 (4) I}{2(R/4)} = 16\left(\frac{\mu_0 I}{2R}\right) = 16B_1$$

In general, when a wire of fixed length is bent into n turns, $B_n = n^2 B_1$. For $n = 4$, $B_2/B_1 = 4^2 = 16$.

BOSONANCE

Practice Question 3

NEET Single Correct • Ampere's Law in Thick Conductor

A solid cylindrical wire of radius $R = 4.0\,\text{mm}$ carries a steady current $I = 10\,\text{A}$ uniformly distributed across its cross section. The ratio of the magnetic field at a radial distance $r_1 = 2.0\,\text{mm}$ to that at $r_2 = 8.0\,\text{mm}$ from the axis of the cylinder is:

(A) $1:1$

(B) $1:2$

(C) $2:1$

(D) $1:4$

Correct Answer: (A) $1:1$

Detailed Reasoning:
Here the wire radius is $R = 4\,\text{mm}$.
• Point 1 is **inside** the wire: $r_1 = 2\,\text{mm} < R$.

$$B(r_1) = \frac{\mu_0 I r_1}{2\pi R^2} = \frac{\mu_0 I(2)}{2\pi(4^2)} = \frac{\mu_0 I}{2\pi} \times \frac{2}{16} = \frac{\mu_0 I}{2\pi} \times \frac{1}{8}$$

• Point 2 is **outside** the wire: $r_2 = 8\,\text{mm} > R$.

$$B(r_2) = \frac{\mu_0 I}{2\pi r_2} = \frac{\mu_0 I}{2\pi(8)} = \frac{\mu_0 I}{2\pi} \times \frac{1}{8}$$

Taking the ratio:

$$\frac{B(r_1)}{B(r_2)} = \frac{\frac{\mu_0 I}{16\pi}}{\frac{\mu_0 I}{16\pi}} = 1:1$$

BOSONANCE

Practice Question 4

NEET Single Correct • Galvanometer Shunts

A moving coil galvanometer has a coil resistance of $99\,\Omega$ and gives full-scale deflection for a current of $10\,\mathrm{mA}$. The value of shunt resistance required to convert this galvanometer into an ammeter reading up to $1.0\,\mathrm{A}$ is:

(A) $0.1\,\Omega$

(B) $1.0\,\Omega$

(C) $9.9\,\Omega$

(D) $0.01\,\Omega$

Correct Answer: (B) $1.0\,\Omega$

Detailed Reasoning:
Given: Galvanometer resistance $R_G = 99\,\Omega$, full-scale current $I_g = 10\,\mathrm{mA} = 0.010\,\mathrm{A}$, total target current $I = 1.0\,\mathrm{A}$.
Current passing through the parallel shunt:

$$I_s = I - I_g = 1.0\,\mathrm{A} - 0.010\,\mathrm{A} = 0.99\,\mathrm{A}$$

Equating potential drops across the parallel branches:

$$r_s = \frac{I_g R_G}{I - I_g} = \frac{0.010 \times 99}{0.99} = \frac{0.99}{0.99} = 1.0\,\Omega$$

BOSONANCE

Practice Question 5

NEET Assertion & Reason Pattern

Assertion (A): When a charged particle moves through a static uniform magnetic field, its kinetic energy remains constant at all times.
Reason (R): The magnetic force on a moving charged particle acts perpendicular to its instantaneous velocity vector, performing zero mechanical work.

(A) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true, but (R) is NOT the correct explanation of (A).

(C) (A) is true, but (R) is false.

(D) (A) is false, but (R) is true.

Correct Answer: (A)

Detailed Reasoning:
The magnetic force is $\mathbf{F}_m = q(\mathbf{v} \times \mathbf{B})$. Because the vector cross product produces a vector strictly normal to both constituents, $\mathbf{F}_m \perp \mathbf{v}$.
Power delivered is $P = \mathbf{F}_m \cdot \mathbf{v} = 0$.
From the Work-Energy Theorem, $W = \Delta K = \int P\,dt = 0 \implies K = \text{constant}$.
Thus, Assertion (A) is true, Reason (R) is true, and (R) correctly explains (A).

BOSONANCE

Practice Question 6

NEET Statement Evaluation Pattern

Examine the following statements regarding a Moving Coil Galvanometer:
Statement I: A soft iron core is placed inside the coil to make the magnetic field radial and increase its magnetic field strength.
Statement II: Increasing the current sensitivity of a galvanometer by doubling the number of turns always results in doubling its voltage sensitivity.
Choose the correct alternative:

(A) Statement I is correct, but Statement II is incorrect.

(B) Statement I is incorrect, but Statement II is correct.

(C) Both Statement I and Statement II are correct.

(D) Both Statement I and Statement II are incorrect.

Correct Answer: (A)

Detailed Reasoning:
Statement I is correct: Soft iron has very high magnetic permeability, which intensifies the field and makes it radial so that $\sin\theta = 1$ across all deflections.
Statement II is incorrect: Current sensitivity is $\frac{\phi}{I} = \frac{NAB}{k}$. If $N \to 2N$, current sensitivity doubles. However, doubling N doubles the wire length, which doubles the resistance $R \to 2R$. Consequently, voltage sensitivity $V_s = \frac{NAB}{kR} \to \frac{(2N)AB}{k(2R)} = \frac{NAB}{kR}$ remains unchanged!

BOSONANCE

Practice Question 7

NEET Single Correct • Parallel Wires

Two long, straight parallel wires separated by a distance of $4.0\,\text{cm}$ carry steady currents of $8.0\,\text{A}$ and $5.0\,\text{A}$ in the same direction. The magnitude of the attractive force experienced by a $10\,\text{cm}$ section of either wire is:

(A) $2.0 \times 10^{-5}\,\text{N}$

(B) $4.0 \times 10^{-5}\,\text{N}$

(C) $1.0 \times 10^{-4}\,\text{N}$

(D) $2.5 \times 10^{-6}\,\text{N}$

Correct Answer: (A) $2.0 \times 10^{-5}\,\text{N}$

Detailed Reasoning:
The force on a section of length L is:

$$F = \frac{\mu_0 I_1 I_2}{2\pi d} L$$

Given: $I_1 = 8.0\,\text{A}$, $I_2 = 5.0\,\text{A}$, $d = 4.0\,\text{cm} = 0.04\,\text{m}$, $L = 10\,\text{cm} = 0.10\,\text{m}$.

$$F = \frac{(4\pi \times 10^{-7}) \times 8.0 \times 5.0}{2\pi \times 0.04} \times 0.10 = \frac{2 \times 10^{-7} \times 40}{0.04} \times 0.10 = \frac{80 \times 10^{-7}}{0.04} \times 0.10$$

$$F = (2000 \times 10^{-7}) \times 0.10 = 2.0 \times 10^{-4} \times 0.10 = 2.0 \times 10^{-5}\,\text{N}$$

BOSONANCE

Practice Question 8

NEET Single Correct • Axial Magnetic Field

At what distance along the axis of a circular current-carrying loop of radius R is the magnetic field equal to $\frac{1}{8}$ of its value at the center of the loop?

(A) $x = \sqrt{2}\,R$

(B) $x = \sqrt{3}\,R$

(C) $x = 2\,R$

(D) $x = 3\,R$

Correct Answer: (B) $x = \sqrt{3}\,R$

Detailed Reasoning:
The field at distance x on the axis is:

$$B_{\text{axial}} = \frac{\mu_0 I R^2}{2(x^2 + R^2)^{3/2}}$$

The field at the center is $B_0 = \frac{\mu_0 I}{2R}$.
Setting $B_{\text{axial}} = \frac{1}{8}B_0$:

$$\frac{\mu_0 I R^2}{2(x^2 + R^2)^{3/2}} = \frac{1}{8}\left(\frac{\mu_0 I}{2R}\right)$$

Canceling common terms $\frac{\mu_0 I}{2}$:

$$\frac{R^2}{(x^2 + R^2)^{3/2}} = \frac{1}{8R} \implies (x^2 + R^2)^{3/2} = 8R^3 = (2R)^3$$

Taking the $2/3$ power of both sides:

$$x^2 + R^2 = (2R)^2 = 4R^2 \implies x^2 = 3R^2 \implies x = \sqrt{3}\,R$$

BOSONANCE

15. Common Pitfalls & NEET Revision Checklist

Top 5 Conceptual Mistakes Made by Students in NEET UG

1. **Confusing the Two Right-Hand Rules:**
   • *For a straight wire:* Thumb = Current, Curled Fingers = Circular Magnetic Field.
   • *For a circular loop/solenoid:* Curled Fingers = Current, Extended Thumb = Axial Magnetic Field / Magnetic Moment.
2. **Forgetting Negative Signs on Electrons:**
   Applying the right-hand rule gives the direction of $\mathbf{v} \times \mathbf{B}$. For an electron ($q = -e$), the actual Lorentz force is directed **opposite** to $\mathbf{v} \times \mathbf{B}$.
3. **Assuming Current Sensitivity Equivalence with Voltage Sensitivity:**
   Remember that $V_s = I_s / R_G$. Increasing turns increases I_s but also increases coil resistance R_G.

4. **Neglecting Field at the Center of Thick Conductors:**
   Remember that inside a solid conductor carrying uniform current, $B = 0$ at the center axis ($r = 0$) and rises linearly ($B \propto r$) up to the surface. It does not drop off from the center!
5. **Force on Curved Wires in Uniform Fields:**
   Do not perform complex line integrals on semicircles or zig-zag wires in uniform $\mathbf{B}$. The effective vector length is simply the straight line displacement $\mathbf{L}'$ connecting the endpoints: $\mathbf{F} = I(\mathbf{L}' \times \mathbf{B})$.

Final 2-Minute NEET Checklist

- Magnetic force never changes speed or kinetic energy of a free charge ($W = 0$).
- Cyclotron period $T = 2\pi m/(qB)$ is independent of speed and radius.
- Biot-Savart permeability: $\mu_0 = 4\pi \times 10^{-7}\ \mathrm{T \cdot m/A}$; $\mu_0 \varepsilon_0 = 1/c^2$.
- Parallel currents attract; antiparallel repel ($f = \mu_0 I_1 I_2/(2\pi d)$).
- Solenoid field: $B = \mu_0 n I$ (center) and $B = \frac{1}{2}\mu_0 n I$ (ends).
- Ammeter = Galvanometer with tiny shunt in parallel ($r_s = I_g R_G/(I - I_g)$).
- Voltmeter = Galvanometer with large resistor in series ($R = V/I_g - R_G$).

BOSONANCE